Wigner semicircle law via Stieltjes transforms + Stein's method

Let $X \in \mathbb{R}^{n \times n}$ be a Wigner matrix, $R(z) = (X - zI)^{-1}$ for $z \in \mathbb{C}^+$.

Recall: The Stieltjes transform of the ESD μ_n of X is

$$m_n(z) = \frac{1}{n} \operatorname{Tr} R(z) = \frac{1}{n} \sum_{i=1}^{n} \frac{1}{\lambda_i - z}$$

For each $z \in \mathbb{C}^+$: $I = (X - zI) R(z) = XR(z) - zR(z)$

Thus $R(z) = \frac{1}{z}(XR(z) - I)$

$$\Rightarrow m_n(z) = \frac{1}{n} \operatorname{Tr} R(z) = \frac{1}{nz} \operatorname{Tr}(XR(z) - I)$$

$$= \frac{1}{nz} \sum_{i=1}^{n} (XR(z))_{ii} - \frac{1}{z}$$

$$= \frac{1}{nz} \sum_{i,j=1}^{n} X_{ij} R(z)_{ji} - \frac{1}{z}.$$

Idea: If $X \sim GOE$, then Gaussian integration-by-parts (Stein's identity) yields a formula for $\mathbb{E} X_{ij} R(z)_{ji}$ that may be used to derive a recursion for $\mathbb{E} m_n(z)$.

Lemma (Stein's identity): Let $\xi \sim N(0,1)$. If $f: \mathbb{R} \to \mathbb{R}$ is continuously differentiable, and $|f(x) \cdot e^{-\frac{x^2}{2}}| \to 0$ as $x \to \pm\infty$, then

$$\mathbb{E}[\xi \cdot f(\xi)] = \mathbb{E}[f'(\xi)]$$

Proof: $\mathbb{E}[\xi \cdot f(\xi)] = \int_{-\infty}^{\infty} \underbrace{f(x)}_{u} \cdot \underbrace{x \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx}_{dv}$, $v = -\frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$

$$= -\int_{-\infty}^{\infty} v\,du = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} f'(x)\,dx$$

$$= \mathbb{E}[f'(\xi)]$$

■

Lemma: Let $X \sim GOE \in \mathbb{R}^{n \times n}$, i.e. $X_{ij} = \frac{1}{\sqrt{n}} z_{ij}$ where

$$z_{ij} \sim \begin{cases} \mathcal{N}(0,1) & \text{if } i < j \\ \mathcal{N}(0,2) & \text{if } i = j \end{cases}, \quad z_{ij} = z_{ji}.$$

Then

$$\mathbb{E}[m_n(z)] = -\frac{1}{z} - \frac{1}{z}\mathbb{E}[m_n(z)^2] - \frac{1}{n^2 z}\mathbb{E}[\operatorname{Tr} R(z)^2]$$

Proof: Let $W \in \mathbb{R}^{n \times n}$ be asymmetric, $w_{ij} \overset{iid}{\sim} \mathcal{N}(0,1)$.

Represent $X = \frac{1}{\sqrt{2n}}(W + W^T)$ (Lecture 1).

Let $\Delta_{ij} \in \mathbb{R}^{n \times n}$ have (i,j) entry 1, remaining entries 0

Then:

- $W = \sum_{i,j=1}^{n} w_{ij} \Delta_{ij}$
- $X - zI = \frac{1}{\sqrt{2n}} \left(\sum_{i,j=1}^{n} w_{ij} \Delta_{ij} + w_{ij} \Delta_{ji} \right) - zI$

$$\Rightarrow \frac{\partial}{\partial w_{ij}}(X - zI) = \frac{1}{\sqrt{2n}}(\Delta_{ij} + \Delta_{ji})$$

- If $A(t) \in \mathbb{C}^{n \times n}$ is invertible, entrywise differentiable in t,

$$I = A(t) \cdot A(t)^{-1}$$

$$\Rightarrow 0 = \frac{d}{dt}\left(A(t) \cdot A(t)^{-1}\right)$$

$$= A'(t) \cdot A(t)^{-1} + A(t) \cdot \frac{d}{dt}\left[A(t)^{-1}\right]$$

$$\Rightarrow \frac{d}{dt}\left[A(t)^{-1}\right] = -A(t)^{-1} \cdot A'(t) \cdot A(t)^{-1}$$

Fix all entries of W except w_{ij}, set $X - zI = A(w_{ij})$.

$$\Rightarrow \frac{\partial}{\partial w_{ij}}(X - zI)^{-1} = -(X - zI)^{-1}\left(\frac{1}{\sqrt{2n}}(\Delta_{ij} + \Delta_{ji})\right)(X - zI)^{-1}$$

$$\text{I.e. } \frac{\partial}{\partial w_{ij}} R(z) = -R(z)\left(\frac{1}{\sqrt{2n}}(\Delta_{ij} + \Delta_{ji})\right)R(z)$$

$$\Rightarrow \frac{\partial}{\partial w_{ij}} R(z)_{ji} = \frac{\partial}{\partial w_{ij}} R(z)_{ij} \qquad \left(R(z) = R(z)^T \text{ b/c } X = X^T\right)$$

$$= -e_i^T R(z)\left(\frac{1}{\sqrt{2n}}(e_i e_j^T + e_j e_i^T)\right)R(z)e_j$$

($e_i \in \mathbb{R}^n$ the ith standard basis vector)

$$= -\frac{1}{\sqrt{2n}}\left(R(z)_{ii}R(z)_{jj} + R(z)_{ij}^2\right)$$

By Stein's identity:

$$\mathbb{E}\left[x_{ij} R(z)_{ji}\right] = \mathbb{E}\left[\frac{1}{\sqrt{2n}}(w_{ij} + w_{ji})R(z)_{ji}\right]$$

$$= \frac{1}{\sqrt{2n}} \mathbb{E}\left[\frac{\partial}{\partial w_{ij}} R(z)_{ij} + \frac{\partial}{\partial w_{ij}} R(z)_{ij}\right]$$

$$= -\frac{1}{n} \mathbb{E}\left[R(z)_{ii} R(z)_{jj} + R(z)_{ij}^2\right]$$

5. $$\mathbb{E}[m_n(z)] = -\frac{1}{z} + \frac{1}{n^2} \sum_{i,j=1}^{n} \mathbb{E}[X_{ij} R(z)_{ij}]$$

$$= -\frac{1}{z} - \frac{1}{n^2 z} \sum_{i,j=1}^{n} \mathbb{E}\left[R(z)_{ii} R(z)_{jj} + R(z)_{ij}^2\right]$$

$$= -\frac{1}{z} - \frac{1}{n^2 z} \mathbb{E}\left[(\mathrm{Tr}\, R(z))^2 + \mathrm{Tr}\, R(z)^2 R(z)\right]$$

$$= -\frac{1}{z} - \frac{1}{z} \mathbb{E}[m_n(z)^2] - \frac{1}{n^2 z} \mathbb{E}\left[\mathrm{Tr}\, R(z)^2\right]$$

Idea: Check that $\mathbb{E}[m_n(z)^2] \approx \mathbb{E}[m_n(z)]^2$, $\frac{1}{n^2}\mathbb{E}[\mathrm{Tr}\, R(z)^2] \approx 0$.

Lemma: Suppose $z = \lambda + i\eta \in \mathbb{C}^+$. Then $|\mathrm{Tr}\, R(z)^2| \le \frac{n}{\eta^2}$.

Proof: Let $X = V(\lambda_1, \dots, \lambda_n) V^T$ be spectral decomp. Then

$$R(z)^2 = [(X - zI)^{-1}]^2$$

$$= V\left(\frac{1}{\lambda_1 - z}, \dots, \frac{1}{\lambda_n - z}\right) V^T V\left(\frac{1}{\lambda_1 - z}, \dots, \frac{1}{\lambda_n - z}\right) V^T$$

$$\Rightarrow \mathrm{Tr}\, R(z)^2 = \sum_{i=1}^{n} \frac{1}{(\lambda_i - z)^2}$$

$$\Rightarrow |\mathrm{Tr}\, R(z)^2| \le \sum_{i=1}^{n} \frac{1}{|\lambda_i - z|^2} \le \frac{n}{\eta^2}.$$

Lemma: Suppose $z = \lambda + i\eta \in \mathbb{C}^+$. Then exist constants $C, C', c > 0$ s.t.

$$\mathbb{P}\left[\left|m_n(z) - \mathbb{E}m_n(z)\right| \geq t\right] \leq Ce^{-cn^2\eta^4 t^2}$$

and hence $\left|\mathbb{E}[m_n(z)^2] - \mathbb{E}[m_n(z)]^2\right| \leq \frac{C'}{n\eta^3}$.

Proof: Recall $m(z) = \int \frac{1}{x-z}\, d\mu(x)$, where

$$\left|\frac{\partial}{\partial z}\frac{1}{z-x}\right| = \frac{1}{|z-x|^2} \leq \frac{1}{dist(z, supp(\mu))^2} \leq \frac{1}{\eta^2}.$$

$\Rightarrow x \mapsto \mathrm{Re}\frac{1}{x-z}$, $x \mapsto \mathrm{Im}\frac{1}{x-z}$ are $\frac{1}{\eta^2}$-Lipschitz.

Then $m_n(z)$ is $\frac{C}{n\eta^2}$-Lipschitz function of iid $\mathcal{N}(0,1)$ variables (Lecture 2). By dimension-free Lipschitz concentration:

$$\mathbb{P}\left[\left|m_n(z) - \mathbb{E}m_n(z)\right| \geq t\right]$$

$$\leq \mathbb{P}\left[\left|\int \mathrm{Re}\frac{1}{x-z}\, d\mu_n(z) - \mathbb{E}\int \mathrm{Re}\frac{1}{x-z}\, d\mu_n(z)\right| \geq \frac{t}{2}\right)$$

$$+\mathbb{P}\left[\int \mathrm{Im}\frac{1}{x-z}\, d\mu_n(z) - \mathbb{E}\int \mathrm{Im}\frac{1}{x-z}\, d\mu_n(z)\right| \geq \frac{t}{2}\right) \leq Ce^{-cn^2\eta^4 t^2}.$$

For second statement:

$$\left|\mathbb{E}\, m_n(z)^2 - (\mathbb{E}m_n(z))^2\right| = \left|\mathbb{E}\left(m_n(z) - \mathbb{E}m_n(z)\right)\left(m_n(z) + \mathbb{E}m_n(z)\right)\right|$$

$$\leq \frac{2}{\eta}\,\mathbb{E}\left|m_n(z) - \mathbb{E}m_n(z)\right| \quad (\text{using } |m_n(z)| \leq \tfrac{1}{\eta}).$$

$$= \frac{2}{\eta} \int_0^\infty \mathbb{P}\left[|m_n(z) - \mathbb{E} m_n(z)| \geq t \right] dt$$

$$\leq \frac{2C}{\eta} \int_0^\infty e^{-cn^2\eta^4 t^2} dt \leq \frac{C'}{n\eta^3}. \quad \blacksquare$$

Proof #3 of Wigner's Thm, when $X \sim GOE$:

By preceding results,

$$\mathbb{E} m_n(z) = -\frac{1}{z} - \frac{1}{z}\left(\mathbb{E} m_n(z)\right)^2 + r_n(z), \quad r_n(z) \to 0.$$

$$\Leftrightarrow \left(\mathbb{E} m_n(z)\right)^2 + z\left(\mathbb{E} m_n(z)\right) + 1 - z r_n = 0$$

So as $n \to \infty$, $\mathbb{E} m_n(z) \to m_{sc}(z)$, root of $m^2 + zm + 1 = 0$ w/ positive imaginary part. From previous lemma + Borel-Cantelli, $m_n(z) - \mathbb{E} m_n(z) \to 0$ a.s. $\forall z \in \mathbb{C}^+$.

So $m_n(z) \to m_{sc}(z)$ $\forall z \in \mathbb{C}^+ \Rightarrow \mu_n \to \mu_{sc}$ weakly a.s. $\blacksquare$

Universality and Lindeberg's method

To illustrate Lindeberg's method, we prove the following CLT:

Thm: Let $x_1, \dots, x_n$ be independent r.v.'s, $\mathbb{E} x_i = 0$, $\mathbb{E} x_i^2 = 1$, $\mathbb{E}|x_i|^3 \leq M$. Let $\varphi : \mathbb{R} \to \mathbb{R}$ have uniformly bounded 3rd derivative. Then

For a universal constant $C>0$,

$$\left|\mathbb{E}\, \varphi\left(\frac{x_1+\cdots+x_n}{\sqrt{n}}\right) - \mathbb{E}_{\xi\sim N(0,1)}\, \varphi(\xi)\right| \le \frac{C M}{\sqrt{n}} \|\varphi'''\|_\infty.$$

Proof: Write $\xi = \frac{z_1+\cdots+z_n}{\sqrt{n}}$, $z_i \overset{iid}{\sim} N(0,1)$ independent of x_i's.

Idea: Swap x_i's one-by-one for z_i's.

Fix any $i \in \{1,\dots,n\}$. Let

$$T^{(i)} = \frac{1}{\sqrt{n}}\left(x_1+\cdots+x_{i-1}+0+z_{i+1}+\cdots+z_n\right)$$

$$S^{(i)} = T^{(i)} + \frac{1}{\sqrt{n}} x_i, \qquad \tilde{S}^{(i)} = T^{(i)} + \frac{1}{\sqrt{n}} z_i.$$

$$\Rightarrow \begin{cases} \varphi(S^{(i)}) = \varphi(T^{(i)}) + \varphi'(T^{(i)})\cdot\frac{x_i}{\sqrt{n}} + \frac{\varphi''(T^{(i)})}{2}\cdot\frac{x_i^2}{n} + \frac{\varphi'''(\tilde{T}^{(i)})}{6}\cdot\frac{x_i^3}{n^{3/2}} \\ \varphi(\tilde{S}^{(i)}) = \varphi(T^{(i)}) + \varphi'(T^{(i)})\cdot\frac{z_i}{\sqrt{n}} + \frac{\varphi''(T^{(i)})}{2}\cdot\frac{z_i^2}{n} + \frac{\varphi'''(\tilde{\tilde{T}}^{(i)})}{6}\cdot\frac{z_i^3}{n^{3/2}} \end{cases}$$

for some (random) $\tilde{T}^{(i)}, \tilde{\tilde{T}}^{(i)} \in \mathbb{R}$. Note: $T^{(i)}$ independent of x_i, z_i

$$\Rightarrow \left|\mathbb{E}\,\varphi(S^{(i)}) - \mathbb{E}\,\varphi(\tilde{S}^{(i)})\right| = \left|\mathbb{E}\left[\frac{\varphi'''(\tilde{T}^{(i)})}{6}\cdot\frac{x_i^3}{n^{3/2}} - \frac{\varphi'''(\tilde{\tilde{T}}^{(i)})}{6}\cdot\frac{z_i^3}{n^{3/2}}\right]\right|$$

$$\le \frac{CM}{n^{3/2}}\cdot\|\varphi'''\|_\infty.$$

Sum over $i = 1,\dots,n$: $\left|\mathbb{E}\,\varphi\left(\frac{x_1+\cdots+x_n}{\sqrt{n}}\right) - \mathbb{E}\,\varphi\left(\frac{z_1+\cdots+z_n}{\sqrt{n}}\right)\right| \le \frac{CM}{\sqrt{n}}\|\varphi'''\|_\infty$. ∎

We now apply this idea to the Stieltjes transform of a Wigner matrix

Lemma: Let $X, \tilde{X} \in \mathbb{R}^{n\times n}$ be two Wigner matrices. Let $m_n, \tilde{m}_n$ be Stieltjes transforms of their ESDs. Then for a constant $C>0$, any $z=\lambda+i\eta \in \mathbb{C}^+$,

$$|\mathbb{E} m_n(z) - \mathbb{E}\tilde{m}_n(z)| \le C\left(\frac{1}{\sqrt{n}\eta^4} + \frac{1}{n\eta^3}\right).$$

Proof: ① Swap off-diagonal: Order $\{(i,j): i\le j\}$ arbitrarily. Set:

$Y^{(i,j)} \in \mathbb{R}^{n\times n}$ symmetric matrix w/ entries

$$Y^{(i,j)}_{kl} = \begin{cases} X_{kl} & \text{if } (k,l) \text{ precedes } (i,j) \\ \tilde{X}_{kl} & \text{if } (k,l) \text{ follows } (i,j) \\ 0 & \text{if } (k,l)=(i,j). \end{cases}$$

$$X^{(i,j)} = Y^{(i,j)} + X_{ij}(\Delta_{ij} + \Delta_{ji}) \quad (\text{replace } 0 \text{ by } X_{ij})$$

$$\tilde{X}^{(i,j)} = Y^{(i,j)} + \tilde{X}_{ij}(\Delta_{ij} + \Delta_{ji}) \quad (\text{replace } 0 \text{ by } \tilde{X}_{ij}).$$

Fix $z=\lambda+i\eta \in \mathbb{C}^+$, fix (i,j), let

$$R = (X^{(i,j)} - zI)^{-1}, \quad m_n^{(i,j)} = \frac{1}{n}\operatorname{Tr} R$$

$$\tilde{R} = (\tilde{X}^{(i,j)} - zI)^{-1}, \quad \tilde{m}_n^{(i,j)} = \frac{1}{n}\operatorname{Tr}\tilde{R}$$

$$S = (Y^{(i,j)} - zI)^{-1}$$

Key identity: $A^{-1} - B^{-1} = A^{-1}\underbrace{(B-A)}_{:=\Delta}B^{-1}$

Then $A^{-1} = B^{-1} + A^{-1}\Delta B^{-1}$, so iterating:

$$A^{-1} = B^{-1} + B^{-1}\Delta B^{-1} + A^{-1}\Delta B^{-1}\Delta B^{-1}$$

$$= B^{-1} + B^{-1}\Delta B^{-1} + B^{-1}\Delta B^{-1}\Delta B^{-1} + A^{-1}\Delta B^{-1}\Delta B^{-1}\Delta B^{-1}$$

$$\Rightarrow m_n^{(ij)} = \frac{1}{n}\operatorname{Tr} R$$

$$= \frac{1}{n}\operatorname{Tr}\left(S + S\Delta S + S\Delta S\Delta S + R\Delta S\Delta S\Delta S\right)$$

where $\Delta = Y^{(ij)} - X^{(ij)} = -x_{ij}\left(\Delta_{ij} + \Delta_{ji}\right)$.

$$\tilde{m}_n^{(ij)} = \frac{1}{n}\operatorname{Tr}\left(S + S\tilde{\Delta}S + S\tilde{\Delta}S\tilde{\Delta}S + \tilde{R}\tilde{\Delta}S\tilde{\Delta}S\tilde{\Delta}S\right)$$

where $\tilde{\Delta} = -\tilde{x}_{ij}\left(\Delta_{ij} + \Delta_{ji}\right)$.

Note: S is independent of $\Delta, \tilde{\Delta}$, and $x_{ij}, \tilde{x}_{ij}$ have matching 1st & 2nd moments.

$$\Rightarrow \mathbb{E} m_n^{(ij)} - \mathbb{E}\tilde{m}_n^{(ij)} = \mathbb{E}\Big[\underbrace{\frac{1}{n}\operatorname{Tr} R\Delta S\Delta S\Delta S}_{(*)} - \underbrace{\frac{1}{n}\operatorname{Tr}\tilde{R}\tilde{\Delta}S\tilde{\Delta}S\tilde{\Delta}S}_{(**)}\Big]$$

$$(*) = \frac{x_{ij}^3}{n}\sum_{k=1}^{n}\left[R\left(\Delta_{ij}+\Delta_{ji}\right)S\left(\Delta_{ij}+\Delta_{ji}\right)S\left(\Delta_{ij}+\Delta_{ji}\right)S\right]_{kk}$$

Note: $\sum_{k=1}^{n}\left[R\Delta_{ab}S\Delta_{cd}S\Delta_{ef}S\right]_{kk}$

$$= \sum_{k=1}^{n} R_{ka}S_{bc}S_{de}S_{fk} = (SR)_{fa}S_{bc}S_{de}.$$

Apply $\|S\|_{op}, \|R\|_{op} \le \frac{1}{\eta}$. (For any $A = V\begin{pmatrix}\lambda_1 & & \\ & \ddots & \\ & & \lambda_n\end{pmatrix}V^T \in \mathbb{R}^{n\times n}$,

$$\|(A - zI)^{-1}\|_{op} = \left\| V \begin{pmatrix} \frac{1}{\lambda_1 - z} & & \\ & \ddots & \\ & & \frac{1}{\lambda_n - z} \end{pmatrix} V^T \right\|_{op} = \max_{i=1}^{n} \frac{1}{|\lambda_i - z|} \le \frac{1}{\eta}.$$

$$\Rightarrow |(SR)_{Ea} S_{bc} S_{dc}| \le \|SR\|_{op} \|S\|_{op} \|S\|_{op} \le \frac{1}{\eta^4}.$$

So $\mathbb{E}|(*)| \le \frac{1}{n\eta^4} \mathbb{E}|X_{ij}|^3 \le \frac{C}{n^{5/2}\eta^4}$

Same for $(**) \Rightarrow \left| \mathbb{E} m_n^{(ij)} - \mathbb{E} \tilde{m}_n^{(ij)} \right| \le \frac{C}{n^{5/2}\eta^4}.$

② Swap diagonal: Set $Y^{(i)} \in \mathbb{R}^{n \times n}$ s.t. $Y^{(i)}_{kk} = \tilde{X}^{(i)}_{kk}$ $\forall k \in [n]$, and

$$Y^{(i)}_{kk} = \begin{cases} X_{kk} & \text{if } k < i \\ \tilde{X}_{kk} & \text{if } k > i \\ 0 & \text{if } k = i \end{cases}$$

$$X^{(i)} = Y^{(i)} + X_{ii} \Delta_{ii}, \quad \tilde{X}^{(i)} = Y^{(i)} + \tilde{X}_{ii} \Delta_{ii}$$

Fix $z = \lambda + i\eta \in \mathbb{C}^+$, fix $i \in \{1, \dots, n\}$, let

$$R = (X^{(i)} - zI)^{-1}, \qquad m_n^{(i)} = \frac{1}{n} \operatorname{Tr} R$$

$$\tilde{R} = (\tilde{X}^{(i)} - zI)^{-1}, \qquad \tilde{m}_n^{(i)} = \frac{1}{n} \operatorname{Tr} \tilde{R}$$

$$S = (Y^{(i)} - zI)^{-1}$$

$$\Rightarrow m_n^{(i)} = \frac{1}{n} \operatorname{Tr} S\Delta S + R\Delta S\Delta S, \qquad \Delta = -X_{ii} \Delta_{ii}$$

$$\tilde{m}_n^{(i)} = \frac{1}{n} \operatorname{Tr} S\tilde{\Delta} S + \tilde{R}\tilde{\Delta} S\tilde{\Delta} S, \qquad \tilde{\Delta} = -\tilde{X}_{ii} \Delta_{ii}$$

Since 1st moments of $X_{ii}, \tilde{X}_{ii}$ match (not necessarily 2nd moments),

$$|\mathbb{E} m_n^{(i)} - \mathbb{E}\tilde{m}_n^{(i)}| = \left|\mathbb{E}\frac{1}{n}\mathrm{Tr}\, R^{(i)}S^{(i)}S - \frac{1}{n}\mathrm{Tr}\,\tilde{R}^{(i)}\tilde{S}^{(i)}\tilde{S}\right|$$

$$= \left|\mathbb{E}\frac{X_{ii}^2}{n}\sum_{k=1}^{n} R^{(i)}_{ki}S^{(i)}_{ii}S^{(i)}_{ik} - \mathbb{E}\frac{\tilde{X}_{ii}^2}{n}\sum_{k=1}^{n}\tilde{R}^{(i)}_{ki}S^{(i)}_{ii}S^{(i)}_{ik}\right|$$

$$\le \frac{1}{n\eta^3}\mathbb{E}|X_{ii}|^2 + \frac{1}{n\eta^3}\mathbb{E}|\tilde{X}_{ii}|^2 \le \frac{C}{n^2\eta^3}.$$

So: $|\mathbb{E} m_n(z) - \mathbb{E}\tilde{m}_n(z)|$

$$\le \left|\sum_{i\neq j}\mathbb{E} m_n^{(ij)} - \mathbb{E}\tilde{m}_n^{(ij)} + \sum_{i=1}^{n}\mathbb{E} m_n^{(i)} - \mathbb{E}\tilde{m}_n^{(i)}\right|$$

$$\le \binom{n}{2}\cdot\frac{C}{n^{5/2}\eta^4} + n\cdot\frac{C}{n^2\eta^3} \le C'\left(\frac{1}{\sqrt{n}\eta^4} + \frac{1}{n\eta^3}\right).$$ ∎

Proof #3 of Wigner's Thm: Let $\tilde{X}\sim GOE$. From the theorem for GOE, $\mathbb{E}\tilde{m}_n(z)\to m_{sc}(z)$ as $n\to\infty$, $\forall z\in\mathbb{C}^+$

From the above lemma, $\mathbb{E} m_n(z) - \mathbb{E}\tilde{m}_n(z)\to 0$ as $n\to\infty$.

Let $X_{ij} = \frac{1}{\sqrt{n}}z_{ij}$. If $\{z_{ij}\}_{i\le j}, \{\frac{1}{\sqrt{2}}z_{ii}\}\overset{i.i.d}{\sim}\nu$ satisfy dim.-free Lipschitz concentration, then as in GOE we have

$$\mathbb{P}\left[|m_n(z) - \mathbb{E} m_n(z)|\ge t\right] \le Ce^{-cn^2\eta^4 t^2}$$

For general Wigner matrix: Write

$$X_1 = (X_{11}, X_{12}, \ldots, X_{1n})$$

$$X_1 = (x_{11}, x_{12}, \dots, x_{1n})$$
$$\vdots$$
$$X_n = (x_{nn})$$

Consider Stieltjes transform as a function $m_n(z) = F(x_1, \dots x_n)$.

Note: $|F(x_1, \dots x_n) - F(\tilde{x}_1, x_2, \dots x_n)|$

$$= \left| \frac{1}{n} \operatorname{Tr}(X - zI)^{-1} - \frac{1}{n} \operatorname{Tr}(\tilde{X} - zI)^{-1} \right|$$

$\tilde{X}$ replaces 1st row/column.

Here $X - \tilde{X}$ is rank 2. By eigenvalue interlacing + same argument as Lecture 3,

$$|F(x_1, \dots x_n) - F(\tilde{x}_1, x_2, \dots x_n)| \le \frac{2v}{n\eta}.$$

Similarly $|F(x_1, \dots x_i, \dots x_n) - F(x_1, \dots \tilde{x}_i, \dots x_n)| \le \frac{2v}{n\eta} \quad \forall i = 1, \dots n.$

Since $x_1, x_2, \dots x_n$ are independent, by McDiarmid's bounded differences inequality,

$$\mathbb{P}[|m_n(z) - \mathbb{E}m_n(z)| \ge t] \le Ce^{-cn\eta^2 t^2}.$$

Then by Borel-Cantelli, $m_n(z) - \mathbb{E}m_n(z) \to 0$ a.s.

Thus: $m_n(z) \to m_{sc}(z)$ a.s. $\forall z \in \mathbb{C}^+ \Rightarrow \mu_n \to \mu_{sc}$ weakly a.s. ∎

Problem 4: Let $Y = A + X \in \mathbb{R}^{n \times n}$ where

- $X \sim GOE$
- $A \in \mathbb{R}^{n \times n}$ is deterministic, symmetric, $\|A\|_{op} \leq C$ a constant, and ESD of A converges weakly to μ_A as $n \to \infty$.

Thus: $\frac{1}{n} \operatorname{Tr}(A - zI)^{-1} \to m_A(z) := \int \frac{1}{x - z} \, d\mu_A(x) \quad \forall z \in \mathbb{C}^+$.

(a) Let $R(z) := (Y - zI)^{-1}$. Show

$$\mathbb{E}[XR(z)] = -\mathbb{E}[m_n(z) R(z)] - \frac{1}{n} \mathbb{E}[R(z)^2],$$

$$= -\mathbb{E} m_n(z) \cdot \mathbb{E} R(z) + r_n, \text{ where } \|r_n\|_{op} \to 0 \text{ as } n \to \infty.$$

(b) Suppose, for each $z \in \mathbb{C}^+$, there exists a unique value $m(z) \in \mathbb{C}^+$ solving the equation

$$m(z) = m_A(z + m(z)),$$

and $m(z)$ is the Stieltjes transform of a prob. measure μ. Show

$$\lim_{n \to \infty} \mathbb{E} m_n(z) = m(z) \quad \forall z \in \mathbb{C}^+$$

$$\lim_{n \to \infty} |m_n(z) - \mathbb{E} m_n(z)| = 0 \quad \text{a.s.} \quad \forall z \in \mathbb{C}^+.$$

Hence ESD of Y converges weakly a.s. to μ.

[μ is the <u>additive free convolution</u> of μ_A and the semicircle law.]